Jiale Chen

+86-137-1610-9390 | 2960452187@qq.com | Beijing, China

Homepage | GitHub | Google Scholar | ORCID

Education

- **Beijing University of Technology** *2023 – 2027*
  B.Eng. in Electronic Science and Technology Beijing, China
  - Overall Average: 88.7/100; Junior-Year Weighted Average: 93.12/100
  - Selected Coursework: Digital Integrated Circuit Design (95), Microcontroller Systems (97), Deep Learning (98), Control Systems (99), RF Integrated Circuit Design (97), Advanced Mathematics (100).

Research Experience

- **FPGA-Based Remote Laboratory Platform** *Mar 2025 – Aug 2025*
  Advisor: Prof. Sujuan Liu | Beijing University of Technology []
  - Designed and implemented a remote FPGA laboratory platform integrating FPGA, Raspberry Pi 5, and STM32 for end-to-end remote hardware experimentation and debugging.
  - Developed a web-based control system supporting remote JTAG programming, logic analysis, virtual oscilloscope functions, digital-twin visualization, and real-time camera monitoring.
  - Implemented UART-based FPGA bitstream transfer at 115200 baud, achieving 10–20 s remote deployment latency and reliable hardware configuration.
  - Built a real-time signal acquisition and debugging framework using FPGA ADC/DAC modules and protocol-level analysis for waveform visualization and system validation.
- **Efficient Adaptation and Deployment of Multimodal Large Language Models** *Mar 2026 – Present*
  Advisor: Hongyu Zhao | Shanghai Jiao Tong University []
  - Developed **MedFusion**, a shared LoRA adaptation framework for multi-source multimodal learning, using fused image-query representations for adapter grouping and dynamic routing.
  - Investigated parameter-efficient adaptation and low-bit quantization for multimodal large language models, focusing on QLoRA, memory-efficient fine-tuning, and efficient model deployment.
  - Developed **MedChain**, an evidence-preservation framework for medical LLM inference, integrating privacy-preserving on-chain/off-chain anchoring with tamper verification for inference records.
- **Communication-Efficient Model Compression for FedDyn in Federated Learning** *Dec 2025 – Mar 2026*
  Advisor: Prof. Ruhui Ma | Shanghai Jiao Tong University []
  - Investigated communication-efficiency bottlenecks in FedDyn under heterogeneous federated learning settings and analyzed the trade-off between transmitted parameter volume and convergence stability.
  - Developed a FedDyn-aware parameter-importance strategy for selective communication, reducing transmission overhead while maintaining model performance.

Industry Experience

- **Glenfly** [][] *Jun 2026 – Aug 2026*
  Software Testing Intern Shanghai, China
  - Conducted functional, stress, stability, and compatibility testing on Glenfly GPUs across Windows and Linux, covering OpenGL, OpenCL, Direct3D, and system-level workloads.
  - Performed repeated GPU workload validation and failure reproduction to evaluate driver compatibility, system stability, and platform behavior under intensive workloads.
  - Developed an automated desktop-folder detection pipeline based on DexiNed and SAM to support repetitive validation workflows across 1080p, 2K, and 4K displays with different DPI settings.

- **Lenovo [][]** *Jul 2025 – Sep 2025*
  Embedded Algorithm Engineer Intern Beijing, China
  - Developed Stable Diffusion 3 (SD3) inference pipelines for text-to-image, image-to-image, and inpainting, integrating SAM/SAM2 for segmentation and mask-guided generation.
  - Built a large-scale data-processing pipeline for millions of image samples and trained ControlNet on image-mask-prompt triplets for segmentation-guided controllable generation.
  - Converted the SD3 + ControlNet inference pipeline from PyTorch to OpenVINO and deployed it on Intel GPUs, completing end-to-end inference validation and hardware-accelerated demo deployment.
- **Cisco []** *Jan 2026 – Feb 2026*
  Technical Engineer Intern Beijing, China
  - Investigated AI data-center networking for distributed GPU training, focusing on RDMA, RoCEv2, ECN/PFC congestion control, and Leaf-Spine network architectures.
  - Analyzed switch architectures and high-performance network designs, and supported validation of network switches and optical modules under data-center scenarios.
  - Configured and validated enterprise routing and switching environments using BGP, OSPF, VLAN, STP, and ACL for protocol verification and troubleshooting.

Publications

[1] **Jiale Chen**.
Communication-Efficient Compression Method for FedDyn Federated Learning Models. []
IEEE 4th International Conference on Control, Electronics and Computer Technology (ICCECT), 2026. Published in IEEE Xplore.

[2] Hongyu Zhao*, **Jiale Chen***, Yihong Chen*.
MedFusion: Shared LoRA Adaptation for Multi-Source Medical Image Classification.
Manuscript in preparation for submission to *IEEE Transactions on Multimedia (TMM)*, 2026.

* Equal contribution.

Honors & Awards

- **Second Prize, National Undergraduate IC Innovation and Entrepreneurship Competition** *Aug 2025*
  Talent Exchange Center, Ministry of Industry and Information Technology (MIIT), China []
- **First Prize, Beijing Integrated Circuit Design Competition (Digital Track)** *May 2026*
  Beijing Municipal Education Commission
- **Third Prize, Chinese Collegiate Computing Competition (National Final)** *Jul 2026*
  Organizing Committee of the Chinese Collegiate Computing Competition []
- **Second Prize, China Undergraduate Mathematical Contest in Modeling (Beijing Region)** *Nov 2025*
  China Society for Industrial and Applied Mathematics (CSIAM) []

Technical Skills

- **Programming:** C, C++, Python, Verilog, CUDA.
- **Hardware & EDA:** Vivado, Quartus II, ModelSim, Cadence Virtuoso, STM32CubeMX, Keil5.
- **AI & Systems:** PyTorch, Hugging Face Transformers, PEFT, OpenVINO, OpenCV.
- **Development:** Linux, Git, Slurm.
- **English:** IELTS Academic: 7.5 (Listening: 8.0, Reading: 8.5, Writing: 6.5, Speaking: 6.0).